Chapter7: Capacitance

Textbook: Fundamentals of Physics by Halliday, Resnick& Walker [extended 9th edition]

Topics to be covered:

25-2: Capacitance, 25-3: Calculating the Capacitance, 25-4: Capacitors in Parallel and in Series, 25-5: Energy Stored in an Electric Field.

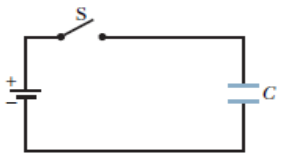
Theory

1. Define Capacitor and Capacitance.
2. Derive the expression for the capacitance of a parallel-plate capacitor.
3. Derive the expression for the capacitance of a Spherical capacitor.
4. Derive the expressions for the equivalent capacitances of series combination (C_S) of capacitors.
5. Derive the expressions for the equivalent capacitances of parallel combination (CP) of capacitors.
6. Find out the expression for the energy stored in an electric field.

Problems

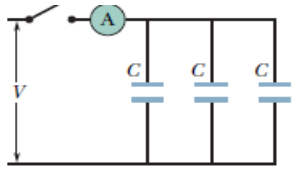
Chapter-25

1. [2] The capacitor in the figure has a capacitance of 25 µF and is initially uncharged. The battery provides a potential difference of 120 V. After switch S is closed,how much charge will pass through it?

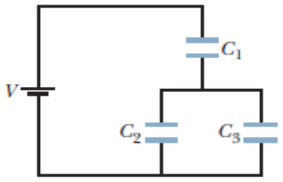


2. [4]The plates of a spherical capacitor have radii 38.0 mm and 40.0 mm.(a) Calculate the capacitance.(b) What must be the plate area of a parallel-plate capacitor with the same plate separation and capacitance?
3. [6] You have two flat metal plates, each of area 1.00m^2, with which to construct a parallel-plate capacitor. If the capacitance of the device is to be 1.00F, what must be the separation between the plates?
4. [5] What is the capacitance of a drop that results when two mercury spheres, each of radius R=2.00 mm, merge?
5. [8] How many 1.00 µF capacitors must be connected in parallel to store a charge of 1.00 C with a potential of 110 V across the capacitors?
6. [72] A potential difference of 300 V is applied to a series connection of two capacitors of capacitances C_1= 2.00 µF andC_2=8.00 µF. What are (a) charge q_1 and (b) potential differenceV_1 on capacitor 1 and (c) q_2 and (d) V_2 on capacitor 2?

7. [9] Each of the uncharged capacitors in the figure has a capacitance of 25.0 µF. A potential difference of V = 4200 V is established when the switch is closed. How many coulombs of charge then pass through meter A?



8. In the figure, find the equivalent capacitance of the combination. Assume that C_1=10.0 µF, C_2= 5.00 µF, and C_3=4.00 µF.



9. [29] What capacitance is required to store an energy of 10 kW.hat a potential difference of 1000 V?
10. [32] A parallel-plate air-filled capacitor having area 40 cm^2 and plate spacing 1.0 mm is charged to a potential difference of 600 V. Find (a) the capacitance, (b) the magnitude of the charge on each plate and (c) the stored energy between the plates.

11. [39] In Figure, C_1= 10.0 µF, C_2= 20.0 µF, and C_3 = 25.0µF. If no capacitor can withstand a potential difference of more than 100 V without failure, what is the magnitude of the maximum potential difference that can exist between points *A* and *B?*

